Persistent Coverage Path Planning for a "Coastal Dome" via Heterogeneous UAV-USV-AUV Collaboration

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Abstract—This research proposes a novel persistent coverage path planning framework to establish a fully functional "Coastal Dome," achieving comprehensive 3D situational awareness (air, surface, and underwater) in coastal environments. The system utilizes a heterogeneous multi-agent network comprising Unmanned Aerial Vehicles (UAVs), Unmanned Surface Vehicles (USVs), and Autonomous Underwater Vehicles (AUVs).

The primary challenge lies in the operational bottlenecks of the agents: the severe energy limitations of the UAVs and the constrained mobility and underwater communication of the AUVs. To overcome this, the study introduces a Dynamic Spatiotemporal Synchronization approach. Rather than relying on static routes, the USV is mathematically formulated as an adaptive operational hub—acting simultaneously as a moving rendezvous landing pad for the UAV's energy recovery and an acoustic relay station for the AUV's data transmission.

The problem is modeled as a multi-objective optimization aimed at minimizing the system's overall Age of Information (AoI) to ensure continuous, persistent coverage, while strictly minimizing energy consumption. By incorporating conditional spatiotemporal constraints, the algorithm guarantees that critical rendezvous events occur before energy depletion. The mathematical model can be solved using either Mixed-Integer Linear Programming (MILP) for exact solutions in discrete grids or metaheuristic approaches like Genetic Algorithms (GA) for scalable, large-area operations, ultimately providing a highly efficient and resilient coastal monitoring system.

Index Terms—Persistent Coverage Path Planning, Heterogeneous Multi-Agent Network (UAV, USV, AUV), Dynamic Spatiotemporal Synchronization, Age of Information Minimization, Multi-Objective Optimization (MILP, Genetic Algorithm).

I. Introduction

PREVIOS research has primarily focused on homogeneous multi-agent frameworks, such as constraint-driven path generation for single-domain USV fleets [1] or dual-agent cooperative search missions balancing coverage efficiency between surface and aerial assets [2], [3]. However, these studies often overlook the tripartite integration of underwater, surface, and aerial domains, failing to address the critical acoustic communication barriers and energy dependency cycles inherent in deep-sea-to-sky synchronization [4], [5], [6]. Existing investigations into cross-domain coordination often prioritize isolated mission parameters, such as individual AUV coverage or restricted 3U target tracking, without synthesizing these heterogeneous constraints into a unified energy-information-decision coupling framework [7], [8], [9]. By leveraging the USV as a centralized communication and energy hub, this framework addresses the synchronization gaps observed in hierarchical multi-agent studies [10], [11]. Building upon decentralized strategies for USV-AUV interaction and swarm-based cooperative coverage, this work extends these methodologies by integrating dynamic rendezvous scheduling into the global objective function [12], [13]. This holistic approach builds upon prior advancements in multi-dimensional optimization for AUV coverage and event-triggered control strategies for heterogeneous maritime search platforms [14], [15]. Previous studies have demonstrated that hierarchical heterogeneous frameworks can mitigate search constraints by decomposing complex maritime tasks into distinct, domain-adaptive sub-operations [16]. Nonetheless, these approaches frequently operate under static environmental assumptions, neglecting the necessity of dynamic, real-time trajectory homotropy and receding horizon control for navigating complex kinodynamic constraints in unpredictable coastal regions [17], [18]. To address these limitations, our framework advances state-of-the-art coordination by implementing an edge-intelligence-driven architecture that facilitates real-time data fusion and adaptive task allocation across disparate domains [19]. By formulating this problem as an Age of Information-aware optimization, our approach ensures that the latency of underwater data collection is minimized while maintaining the operational endurance of airborne assets [20].

II. Where to Get LaTeX Help — User Groups

The following online groups are helpful to beginning and experienced LaTeX users. A search through their archives can provide many answers to common questions.

http://www.latex-community.org/
https://tex.stackexchange.com/

III. Other Resources

See [1]–[5] for resources on formatting math into text and additional help in working with LaTeX.

IV. Text

For some of the remainer of this sample we will use dummy text to fill out paragraphs rather than use live text that may violate a copyright.

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$$x = \sum_{i=0}^{n} 2iQ. \tag{1}$$

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V. Some Common Elements

A. Sections and Subsections

Enumeration of section headings is desirable, but not required. When numbered, please be consistent throughout the article, that is, all headings and all levels of section headings in the article should be enumerated. Primary headings are designated with Roman numerals, secondary with capital letters, tertiary with Arabic numbers; and quaternary with lowercase letters. Reference and Acknowledgment headings are unlike all other section headings in text. They are never enumerated. They are simply primary headings without labels, regardless of whether the other headings in the article are enumerated.

B. Citations to the Bibliography

The coding for the citations is made with the LaTeX \cite command. This will display as: see [1].

For multiple citations code as follows: \cite{ref1,ref2,ref3} which will produce [1]–[3]. For reference ranges that are not consecutive code as \cite{ref1,ref2,ref3,ref9} which will produce [1]–[3], [9]

C. Lists

In this section, we will consider three types of lists: simple unnumbered, numbered, and bulleted. There have been many options added to IEEEtran to enhance the creation of lists. If your lists are more complex than those shown below, please refer to the original "IEEEtran_HOWTO.pdf" for additional options.

A plain unnumbered list:

bare_jrnl.tex
bare_conf.tex
bare_jrnl_compsoc.tex
bare_conf_compsoc.tex
bare_jrnl_comsoc.tex

A simple numbered list:

1) bare_jrnl.tex
2) bare_conf.tex
3) bare_jrnl_compsoc.tex
4) bare_conf_compsoc.tex
5) bare_jrnl_comsoc.tex

A simple bulleted list:

- bare_jrnl.tex
- bare_conf.tex
- bare_jrnl_compsoc.tex
- bare_conf_compsoc.tex
- bare_jrnl_comsoc.tex

D. Figures

Fig. 1 is an example of a floating figure using the graphicx package. Note that \label must occur AFTER (or within) \caption. For figures, \caption should occur after the \includegraphics.

Fig. 2(a) and 2(b) is an example of a double column floating figure using two subfigures. (The subfig.sty package must be loaded for this to work.) The subfigure \label commands are set within each subfloat command, and the \label for the overall figure must come after \caption. \hfil is used as a separator to get equal spacing. The combined width of all the parts of the figure should do not exceed the text width or a line break will occur.



Fig. 1. Simulation results for the network.

TABLE I
An Example of a Table

One	Two
Three	Four

Note that often IEEE papers with multi-part figures do not place the labels within the image itself (using the optional argument to \subfloat[]), but instead will reference/describe all of them (a), (b), etc., within the main caption. Be aware that for subfig.sty to generate the (a), (b), etc., subfigure labels, the optional argument to \subfloat must be present. If a subcaption is not desired, leave its contents blank, e.g.,\subfloat[].

VI. Tables

Note that, for IEEE-style tables, the \caption command should come BEFORE the table. Table captions use title case. Articles (a, an, the), coordinating conjunctions (and, but, for, or, nor), and most short prepositions are lowercase unless they are the first or last word. Table text will default to \footnotesize as the IEEE normally uses this smaller font for tables. The \label must come after \caption as always.

VII. Algorithms

Algorithms should be numbered and include a short title. They are set off from the text with rules above and below the title and after the last line.

Algorithm 1 Weighted Tanimoto ELM.

TRAIN($\mathbf{XT}$)
  select randomly $W \subset \mathbf{X}$
  $N_{\mathbf{t}} \leftarrow |\{i : \mathbf{t}_i = \mathbf{t}\}|$ for $\mathbf{t} = -1, +1$
  $B_i \leftarrow \sqrt{\max(N_{-1}, N_{+1})/N_{\mathbf{t}_i}}$ for $i = 1, ..., N$
  $\hat{\mathbf{H}} \leftarrow B \cdot (\mathbf{X}^T \mathrm{W})/(\not\Vdash\mathbf{X} + \not\Vdash\mathrm{W} - \mathbf{X}^T \mathrm{W})$
  $\beta \leftarrow \left(I/C + \hat{\mathbf{H}}^T \hat{\mathbf{H}}\right)^{-1} (\hat{\mathbf{H}}^T B \cdot \mathbf{T})$
  return W, β

PREDICT($\mathbf{X}$)
  $\mathbf{H} \leftarrow (\mathbf{X}^T \mathrm{W})/(\not\Vdash\mathbf{X} + \not\Vdash\mathrm{W} - \mathbf{X}^T \mathrm{W})$
  return sign($\mathbf{H}\beta$)

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VIII. Mathematical Typography and Why It Matters

Typographical conventions for mathematical formulas have been developed to provide uniformity and clarity of presentation across mathematical texts. This enables the readers of those texts to both understand the author's ideas and to grasp new concepts quickly. While software such as LaTeX and MathType® can produce aesthetically pleasing math when used properly, it is also very easy to misuse the software, potentially resulting in incorrect math display.

IEEE aims to provide authors with the proper guidance on mathematical typesetting style and assist them in writing the best possible article. As such, IEEE has assembled a set of examples of good and bad mathematical typesetting [1]–[5].

Further examples can be found at http://journals.ieeeauthorcenter.ieee.org/wp-content/uploads/sites/7/IEEE-Math-Typesetting-Guide-for-LaTeX-Users.pdf

A. Display Equations

The simple display equation example shown below uses the "equation" environment. To number the equations, use the \label macro to create an identifier for the equation. LaTeX will automatically number the equation for you.

$$x = \sum_{i=0}^{n} 2iQ. \tag{2}$$

is coded as follows:

```
\begin{equation}
\label{deqn_ex1}
x = \sum_{i=0}^{n} 2{i} Q.
\end{equation}
```



(a) (b)

Fig. 2. Dae. Ad quatur autat ut porepel itemoles dolor autem fuga. Bus quia con nessunti as remo di quatus non perum que nimus. (a) Case I. (b) Case II.

To reference this equation in the text use the \ref macro. Please see (2)
is coded as follows:

```
Please see (\ref{deqn_ex1})
```

B. Equation Numbering

Consecutive Numbering: Equations within an article are numbered consecutively from the beginning of the article to the end, i.e., (1), (2), (3), (4), (5), etc. Do not use roman numerals or section numbers for equation numbering.
Appendix Equations: The continuation of consecutively numbered equations is best in the Appendix, but numbering as (A1), (A2), etc., is permissible.

Hyphens and Periods: Hyphens and periods should not be used in equation numbers, i.e., use (1a) rather than (1-a) and (2a) rather than (2.a) for subequations. This should be consistent throughout the article.

C. Multi-Line Equations and Alignment

Here we show several examples of multi-line equations and proper alignments.
A single equation that must break over multiple lines due to length with no specific alignment.

$$\begin{multline}\text{The first line of this example}\\ \text{The second line of this example}\\ \text{The third line of this example}\end{multline} \tag{3}$$

is coded as:

```
\begin{multline}
\text{The first line of this example}\\
\text{The second line of this example}\\
\text{The third line of this example}
\end{multline}
```

A single equation with multiple lines aligned at the = signs

$$\begin{aligned} a &= c+d \\ b &= e+f \end{aligned} \tag{4,5}$$

is coded as:

```
\begin{align}
a &= c+d \\
b &= e+f
\end{align}
```

The align environment can align on multiple points as shown in the following example:

$$\begin{aligned} x &= y & X &= Y & a &= bc \\ x' &= y' & X' &= Y' & a' &= bz \end{aligned} \tag{6,7}$$

is coded as:

```
\begin{align}
x &= y & X &= Y & a &= bc\\
x' &= y' & X' &= Y' &a' &= bz
\end{align}
```

D. Subnumbering

The amsmath package provides a subequations environment to facilitate subnumbering. An example:

$$\begin{aligned} f &= g \\ f' &= g' \\ \mathcal{L}f &= \mathcal{L}g \end{aligned} \tag{8a,8b,8c}$$

is coded as:

```
\begin{subequations}\label{eq:2}
\begin{align}
f&=g \label{eq:2A}\\
f' &=g' \label{eq:2B}\\
```

```
\mathcal{L}f &= \mathcal{L}g \label{eq:2c}
\end{align}
\end{subequations}
```

E. Matrices

There are several useful matrix environments that can save you some keystrokes. See the example coding below and the output.

A simple matrix:

$$\begin{matrix} 0 & 1 \\ 1 & 0 \end{matrix} \tag{9}$$

is coded as:

```
\begin{equation}
\begin{matrix}  0 &  1 \\
1 &  0 \end{matrix}
\end{equation}
```

A matrix with parenthesis

$$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \tag{10}$$

is coded as:

```
\begin{equation}
\begin{pmatrix} 0 & -i \\
 i &  0 \end{pmatrix}
\end{equation}
```

A matrix with square brackets

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \tag{11}$$

is coded as:

```
\begin{equation}
\begin{bmatrix} 0 & -1 \\
1 &  0 \end{bmatrix}
\end{equation}
```

A matrix with curly braces

$$\begin{Bmatrix} 1 & 0 \\ 0 & -1 \end{Bmatrix} \tag{12}$$

is coded as:

```
\begin{equation}
\begin{Bmatrix} 1 &  0 \\
0 & -1 \end{Bmatrix}
\end{equation}
```

A matrix with single verticals

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} \tag{13}$$

is coded as:

```
\begin{equation}
\begin{vmatrix} a &  b \\
c &  d \end{vmatrix}
\end{equation}
```

A matrix with double verticals

$$\begin{Vmatrix} i & 0 \\ 0 & -i \end{Vmatrix} \tag{14}$$

is coded as:

```
\begin{equation}
\begin{Vmatrix} i &  0 \\
0 & -i \end{Vmatrix}
\end{equation}
```

F. Arrays

The array environment allows you some options for matrix-like equations. You will have to manually key the fences, but there are other options for alignment of the columns and for setting horizontal and vertical rules. The argument to array controls alignment and placement of vertical rules.

A simple array

$$\left(\begin{array}{cccc} a+b+c & uv & x-y & 27 \\ a+b & u+v & z & 134 \end{array} \right) \tag{15}$$

is coded as:

```
\begin{equation}
\left(
\begin{array}{cccc}
a+b+c & uv & x-y & 27\\
a+b & u+v & z & 134
\end{array} \right)
\end{equation}
```

A slight variation on this to better align the numbers in the last column

$$\left(\begin{array}{cccc} a+b+c & uv & x-y & 27 \\ a+b & u+v & z & 134 \end{array} \right) \tag{16}$$

is coded as:

```
\begin{equation}
\left(
\begin{array}{cccr}
a+b+c & uv & x-y & 27\\
a+b & u+v & z & 134
\end{array} \right)
\end{equation}
```

An array with vertical and horizontal rules

$$\left(\begin{array}{c|c|c|r} a+b+c & uv & x-y & 27 \\ \hline a+b & u+v & z & 134 \end{array} \right) \tag{17}$$

is coded as:

```
\begin{equation}
\left(
\begin{array}{c|c|c|r}
a+b+c & uv & x-y & 27\\
a+b & u+v & z & 134
\end{array} \right)
\end{equation}
```

Note the argument now has the pipe "|" included to indicate the placement of the vertical rules.

G. Cases Structures

Many times cases can be miscoded using the wrong environment, i.e., array. Using the cases environment will save keystrokes (from not having to type the \left\lbrace) and automatically provide the correct column alignment.

$$z_m(t) = \begin{cases} 1, & \text{if } \beta_m(t) \\ 0, & \text{otherwise.} \end{cases}$$

is coded as follows:

```
\begin{equation*}
{z_m(t)} =
\begin{cases}
1,&{\text{if}}\ {\beta }_m(t),\\
{0,}&{\text{otherwise.}}
\end{cases}
\end{equation*}
```

Note that the "&" is used to mark the tabular alignment. This is important to get proper column alignment. Do not use \quad or other fixed spaces to try and align the columns. Also, note the use of the \text macro for text elements such as "if" and "otherwise."

H. Function Formatting in Equations

Often, there is an easy way to properly format most common functions. Use of the \ in front of the function name will in most cases, provide the correct formatting. When this does not work, the following example provides a solution using the \text macro:

$$d_R^{KM} = \underset{d_l^{KM}}{\arg\min}\{d_1^{KM}, \ldots, d_6^{KM}\}.$$

is coded as follows:

```
\begin{equation*}
 d_{R}^{KM} = \underset {d_{l}^{KM}}
 {\text{arg min}} \{ d_{1}^{KM},
 \ldots,d_{6}^{KM}\}.
\end{equation*}
```

I. Text Acronyms Inside Equations

This example shows where the acronym "MSE" is coded using \text{} to match how it appears in the text.

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$$

```
\begin{equation*}
 \text{MSE} = \frac {1}{n}\sum _{i=1}^{n}
(Y_{i} - \hat {Y_{i}})^{2}
\end{equation*}
```

IX. Conclusion

The conclusion goes here.

Acknowledgments

This should be a simple paragraph before the References to thank those individuals and institutions who have supported your work on this article.

Appendix
Proof of the Zonklar Equations

Use \appendix if you have a single appendix: Do not use \section anymore after \appendix, only \section*. If you have multiple appendixes use \appendices then use \section to start each appendix. You must declare a \section before using any \subsection or using \label (\appendices by itself starts a section numbered zero.)

References Section

You can use a bibliography generated by BibTeX as a .bbl file. BibTeX documentation can be easily obtained at: http://mirror.ctan.org/biblio/bibtex/contrib/doc/ The IEEEtran BibTeX style support page is: http://www.michaelshell.org/tex/ieeetran/bibtex/

Simple References

You can manually copy in the resultant .bbl file and set second argument of \begin to the number of references (used to reserve space for the reference number labels box).

References

[1] Mathematics Into Type. American Mathematical Society. [Online]. Available: https://www.ams.org/arc/styleguide/mit-2.pdf

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Biography Section

If you have an EPS/PDF photo (graphicx package needed), extra braces are needed around the contents of the optional argument to biography to prevent the LaTeX parser from getting confused when it sees the complicated \includegraphics command within an optional argument. (You can create your own custom macro containing the \includegraphics command to make things simpler here.)

If you include a photo:



Michael Shell Use \begin{IEEEbiography} and then for the 1st argument use \includegraphics to declare and link the author photo. Use the author name as the 3rd argument followed by the biography text.

If you will not include a photo:

John Doe Use \begin{IEEEbiographynophoto} and the author name as the argument followed by the biography text.